

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Developed infrastructure services in **C++**, **Go**, and **Python** for **GPU** scheduling, resource allocation, and fault-tolerant workload placement across **AWS**, **GCP**, and **CoreWeave** compute fleets under capacity constraints.
- Operated multi-cloud GPU clusters with **InfiniBand/RoCE** networking and **Slurm**-based HPC scheduling, keeping fleet utilization high through node loss and provider disruptions.
- Debugged production failures spanning compute nodes, **network** dependencies, schedulers, and serving services, driving fixes through implementation, testing, code review, and rollout.
- Built **Prometheus/Grafana** observability and autoscaling for inference serving, correlating compute, **network**, and **storage** telemetry to identify fleet-wide **performance** regressions.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built and operated **Kubernetes**-based **GPU** infrastructure across multiple cloud providers, developing **Go** and **Python** tooling for workload scheduling and fault-tolerant placement.
- Reduced **p99 serving latency** from **420ms to 140ms** across 50K+ daily requests through asynchronous execution redesign, **profiling**, and systems-level performance optimization.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Designed and operated **25+ distributed C++/Java** services for **1M+ users** on **AWS EC2**, **RDS**, and **SQS**, sustaining **99.9% availability** over four years of production ownership.
- **Profiled compute and memory bottlenecks** across **C++/Python** services and optimized batching, scheduling, and **concurrency**, raising throughput **2.4x** and cutting memory usage **27%**.
- Diagnosed production incidents across application, database, **network**, and downstream-service boundaries using logs, metrics, and root-cause analysis; codified fixes into shared standards, cutting escalations **60%**.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus** distributed key-value store with **linearizable reads**, leader election, and **log compaction**; companion VPC control-plane simulator modeling subnet routing.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **Multi-agent** incident-diagnosis system orchestrating tools to collect evidence from logs and metrics, analyze production failures, and synthesize root-cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Model **evaluation** platform with automated regression suites in **CI/CD**; **FastAPI** and **PostgreSQL** tracking quality and latency across deployments.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Go, Python, Java, SQL

Distributed Systems: Scheduling, Fault Tolerance, Resource Allocation, Concurrency, Consensus (Raft), Service Reliability

Infrastructure: Linux, GPU Compute, InfiniBand/RoCE, Slurm, Storage Systems, Observability, Performance Profiling, Debugging

Cloud & DevOps: GCP, AWS, CoreWeave, Kubernetes, Docker, Terraform, CI/CD, Prometheus, Grafana

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer